When Consensus Is Evidence and When It Is Not

Abstract

Expert agreement can be useful evidence, but its meaning depends on who was asked and what they agreed about. This paper examines four questions: how independent the judgments are, whether dissent can succeed on evidence, whether claims can be tested, and how the surveyed group is defined. Climate-consensus surveys illustrate why a percentage must travel with its actual question and denominator. Restricting a survey to relevant experts or expressed opinions can be appropriate; extending its answer to a broader claim requires justification. The paper proposes a reporting standard that makes those distinctions visible without treating disagreement as evidence that the underlying science is false.

1. The Problem With Citing Agreement

A reader cannot personally check every field on which a decision depends. Relevant expert agreement can therefore provide useful evidence. The question is what a particular consensus figure tells us: who agreed, which proposition they endorsed and how their judgments were formed.

The interesting question is therefore not whether consensus counts but under what conditions it counts, and how much. That question has received far less attention than its practical importance warrants, in part because it is usually raised by people with a stake in rejecting a particular consensus, which has made the question itself suspect by association.

The suspicion is understandable and it is a mistake. **A discipline that cannot say when its own appeals to agreement are load-bearing has left its epistemology to be adjudicated by whoever shouts loudest**, and the record shows that this adjudication goes badly in both directions: real consensus is dismissed by people who dislike its implications, and manufactured figures are cited by people who should have checked them.

2. Four Conditions

Expert agreement functions as evidence to the extent that the following hold.

Independence. The value of many assenting judgements depends on those judgements being arrived at separately. A hundred experts who each assessed the evidence independently constitute a hundred data points. A hundred experts who each deferred to the same three reviews do not supply a hundred independent assessments. Their effective evidential contribution depends on the dependence among the reviews, their underlying evidence and what each expert independently checked; it cannot be read off as exactly three or one. Independence is rarely perfect and is often much lower than the raw count suggests, because training, funding, and publication all flow through shared channels.

A reward for successful dissent. A field in which overturning the majority view is the fastest route to a career is a field whose majority view has survived people trying to break it. Where dissent is penalized regardless of its evidential quality, agreement loses one important check. Establishing that this is happening requires evidence about how claims were evaluated; the existence of disagreement alone does not establish it. This condition is about incentive structure rather than about individual virtue, and it is the condition most often assumed rather than checked.

Exposure to test. A claim generating predictions that could have failed and did not is in a different position from a claim compatible with any observation. This is Popper's condition, and it applies to the consensus rather than merely to the claim: what matters is whether the assenting experts would have changed their assent had the predictions failed.

Stability of the reference class. A proportion is a ratio, and a ratio has a denominator. If the denominator is constructed by the same people reporting the ratio, and if the construction is what produces the figure, then the figure measures the construction. This condition is the least discussed and, we will argue, the one that fails most often in public citation.

3. The Reference-Class Problem

Every proportion requires a decision about what to count. In most reporting the decision is invisible, and its invisibility is what allows a constructed figure to be received as a measured one.

The failure mode is not fabrication. It is a sequence of individually defensible narrowings whose product is a number that answers a different question from the one the reader believes was asked.

Consider the abstract structure. A population of N items is surveyed. Some subset expresses a view on the question. Within that subset, some proportion agrees. The reported figure is the proportion within the subset, while the number the reader retains is a proportion of the population. Each step admits a defence: those who expressed no view cannot be counted as disagreeing; those outside the specialty are not qualified. **The defences can be reasonable. The product becomes misleading when a later summary conceals the restricted denominator. If the original paper reports the denominator, that criticism belongs to the summary, not to the act of defining a relevant sample.**

4. The Climate-Agreement Surveys as a Case

We take the two most-cited surveys because they are the most-cited, not because the underlying science is our subject. **Nothing in this section bears on whether anthropogenic warming is occurring.** The question is what the surveys measured.

Cook et al. (2013) examined 11,944 abstracts. Of these, 7,930 took no position on attribution and were set aside. Among the 4,014 that did take a position, 3,896 were classed as endorsing, and 3,896 over 4,014 is the 97.1 percent that circulates. Against the full sample the endorsing abstracts are 32.6 percent. The paper reports both figures.

The coding matters more than either. Cook pooled three categories as endorsement. Category 1 required an explicit, quantified statement that humans were the primary cause. Categories 2 and 3 required only that a paper endorse human-caused warming explicitly without quantifying it, or imply it without saying so. **The 97.1 percent is therefore not a finding that 97.1 percent of papers endorsed the claim that humans caused most warming.** That is a different proposition with a different count.

Legates et al. (2015) applied the narrower criterion directly. They inspected Cook's 64 category-1 abstracts, judged that 23 did not explicitly support the proposition that human activity caused most warming since 1950, and arrived at **41 papers, which is 0.34 percent of the original sample.**

The two counts answer different questions and both are arithmetically correct. Cook asked whether a paper endorsed human-caused warming, including implicitly. Legates asked whether it explicitly supported a quantified claim about most warming since 1950. **The reported proportion depends entirely on which question is being asked, and the headline figure corresponds to the broadest one.**

A later survey avoided the conditional denominator by estimating agreement against the entire search corpus rather than excluding no-position papers from the headline. Lynas, Houlton and Perry (2021) report greater than 99 percent agreement on that basis, which is the methodologically cleaner design and is rarely the number cited.

Doran and Zimmerman (2009) invited 10,257 earth scientists. 3,146 responded. The specialist subgroup was those listing climate science as their field who had published more than half their recent papers on it: 79 people. Of those, 76 answered that temperatures had risen. On the human-contribution question, **75 of the 77 who answered it said yes, which is the 97.4 percent.** The paper reports all of this.

Doran and Zimmerman's 97.4 percent figure rests on 75 affirmative answers among 77 respondents in the relevant specialist subgroup. That denominator belongs to this survey. It is not the sole basis of every published estimate of climate-science agreement; Cook et al. used a different sample and method.

Neither paper concealed anything. Both state their methods. The problem lies in the transmission: a figure computed over a deliberately narrowed reference class entered public discourse as though computed over the whole.

And the cost falls on the claim it was meant to support. A reader who

checks the derivation and finds the denominator moved does not conclude that the reference class was reasonable. He concludes he was handled, and he then discounts the actual evidence, which is substantial and which did not need the survey.

5. What Peer Review Does and Does Not Establish

Peer review is often invoked as though passage through it certified truth. It does not, and its practitioners generally do not claim it does.

Peer review can identify errors, missing citations, unclear methods and even deliberate misrepresentation. Ordinary manuscript review is not necessarily a raw-data audit or an independent replication. Its evidential value depends on what reviewers actually checked and on the expertise and independence they brought to that task. Acceptance should therefore be accompanied by a clear account of data access, code availability and any replication or verification performed.

The replication crisis is the relevant evidence. Large systematic replication efforts in psychology, cancer biology, and economics have all found substantial fractions of published, peer-reviewed findings failing to replicate. Whatever peer review certified in those cases, it was not reproducibility.

Peer review is therefore best understood as a filter for conformity to disciplinary standards, which is valuable and is not the same as a filter for truth. In a healthy field the two correlate strongly, because the standards were built by the evidence. In a field where dissent is punished, the correlation weakens precisely when it is most needed, since conformity to standards is exactly what a manufactured consensus produces.

6. Distinguishing the Manufactured From the Real

The argument so far could be read as licensing generalised suspicion. It does not, and the distinction matters.

Ordinary convergence occurs when independent lines of evidence, gathered by different methods for different purposes, point the same way. The age of the Earth is established by radiometric dating across multiple decay chains, by stellar physics, and by meteoritic evidence, none of which was designed to confirm the others. Convergence of that kind is strong evidence and is not vulnerable to the reference-class problem, because it does not rest on a count of who agrees.

Manufactured consensus requires more than a categorisation decision: every survey needs categories. Evidence of manufacture would include undisclosed selection that changes the target population, selective reporting, coercion, or presenting dependent endorsements as independent confirmation. A transparent estimate of agreement can instead summarize genuine evidential convergence.

The question is whether the reported claim matches the sampling and endorsement rule.

The diagnostic is simple and it can be applied by a non-specialist. **Ask what the denominator is, and ask who chose it.** Then ask whether that reference class fits the question being answered, whether exclusions were disclosed, and how the estimate changes under reasonable alternatives. Selecting a relevant expert subgroup can improve the estimate. An undisclosed change from that subgroup to "all scientists" in a public summary changes the claim.

7. A Reporting Standard

We propose that any published proportion of expert agreement report four quantities: the full population surveyed, the number expressing a view, the number meeting each inclusion criterion applied, and the proportion computed against each of these denominators.

Both surveys examined above would satisfy this standard, since both report the underlying numbers. **The standard is therefore not a criticism of the papers but of the citation practice**, and its value is that it makes the citation practice harder. A figure published alongside its own alternative denominators is more difficult to transmit misleadingly.

Journals could adopt this at negligible cost. It requires no new analysis, only that the numbers already computed be reported together.

8. Objections

That this arms bad-faith critics. It does, and the alternative arms them better. A critic who discovers the denominator problem unaided has been handed both the discrepancy and the evidence that it went unmentioned. Stating it first costs the discrepancy and keeps the credibility.

That non-specialists cannot evaluate primary evidence and must defer. Agreed, and nothing here says otherwise. The conditions in section 2 are conditions on when deference is warranted, not an argument against deferring. A layman who checks the denominator has not evaluated the climate science; he has evaluated whether the number he was given measures what he was told it measures, which is a question he can answer.

That the underlying claim is well supported regardless. Also agreed, and it is the point. The physical evidence for anthropogenic warming does not depend on the survey, which is why citing the survey was a poor decision: it attached a strong case to a weak instrument, and the instrument is what gets attacked.

9. Conclusion

Expert agreement is evidence whose strength depends on how the agreement was reached and measured. A reported percentage should identify its question, surveyed group, exclusions and denominator.

The proposed reporting standard makes those checks easier. It neither treats consensus as proof nor treats a limitation in a survey as a refutation of the science. Its purpose is to let public claims retain the meaning that the underlying evidence actually supports.

References

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VERIFIED against the primary sources, August 2026. Cook figures confirmed: 11,944 / 7,930 no position / 4,014 expressing a position / 3,896 endorsing / 97.1% conditional / 32.6% of full sample / 0.7% rejecting / 0.3% uncertain. Doran-Zimmerman confirmed: 10,257 invited, 3,146 responded, 79 specialists, 76 of 79 on warming, 75 of 77 on human contribution. Legates confirmed: 64 category-1 abstracts inspected, 23 judged not to meet the narrower criterion, 41 remaining, 0.34% of sample. **No formal Cook reply to Legates could be verified; the claim has been removed.** Lynas, Houlton & Perry (2021), Environmental Research Letters 16, 114005, added.

Approximately 2,100 words.

Required claim matching

For each survey, report its sampling unit, eligibility rules, response or position categories, exact endorsement question, denominator and treatment of missing responses. Then quote the public claim being evaluated. A proportion about one

population does not automatically measure another; neither does the difference establish deliberate manufacture. The case analysis needs this paired record before a general conclusion about public misuse.